Weekly Report

Dean Reiter

March 31, 2026

Summary

- Following up data shown last week
- Cosmic vs Moller scattering data
- Detector hit position correlations between **isolated** hits
  - i.e. only showing correlations between a single hit in each detector

Dataset Information

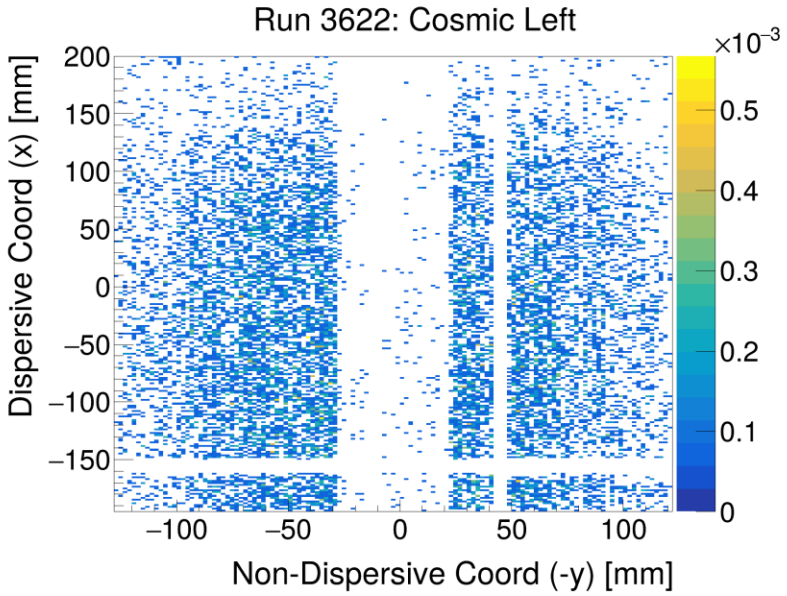
- 1M events each
- GEMs at 3950V

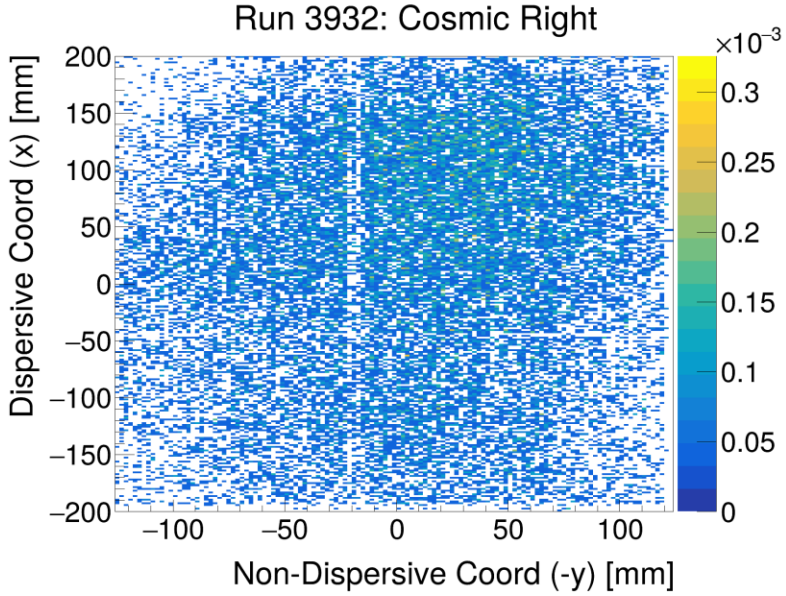
*Many dead strips in left bottom GEM

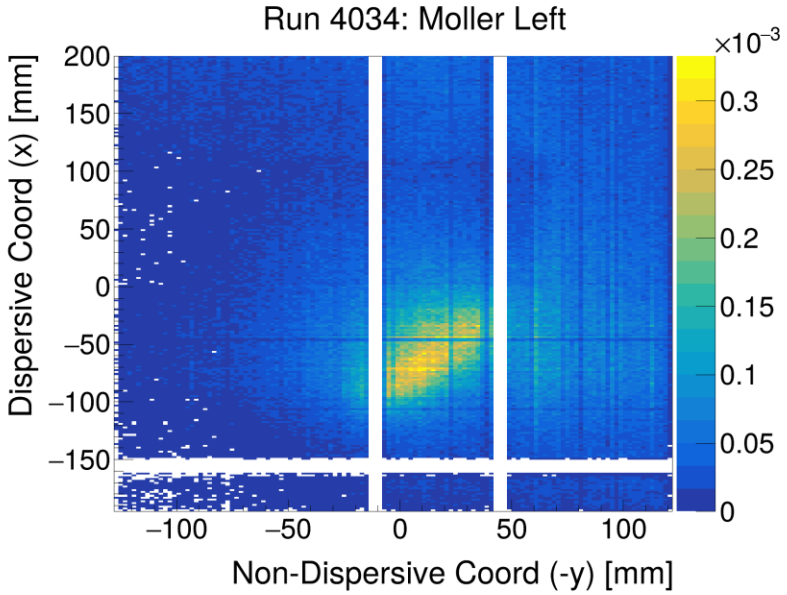
Run	Date	Setup	Readout	Dipole	Beam
3622*	4 Jan 2026	Cosmic	L+Trig		
3932	15 Jan 2026	Cosmic	R+Trig		
4026	21 Jan 2026	e-e-	Full	90/30	15 MeV, 5uA, 13us, 1.5%, 1100Hz
4034	21 Jan 2026	e-e-	Full	12.8/90	``

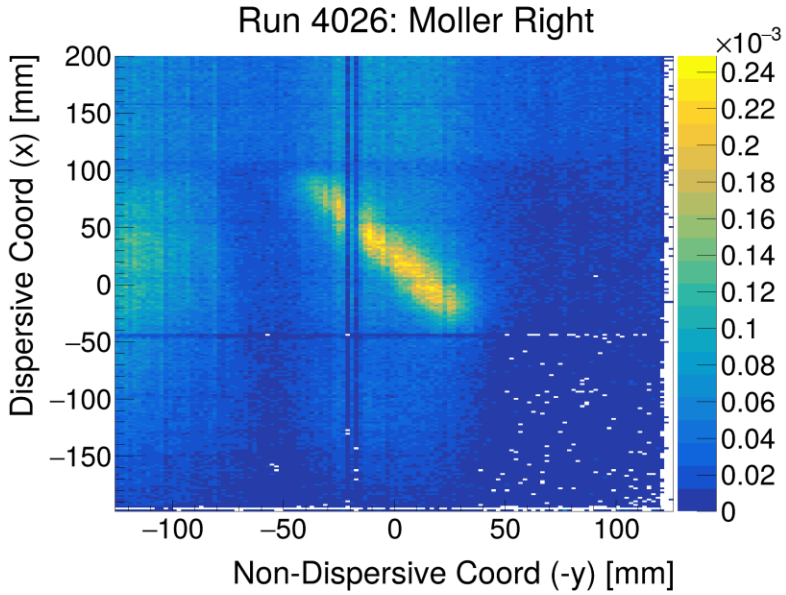
GEM Hit Positions

Bottom GEM Cluster Local Positions

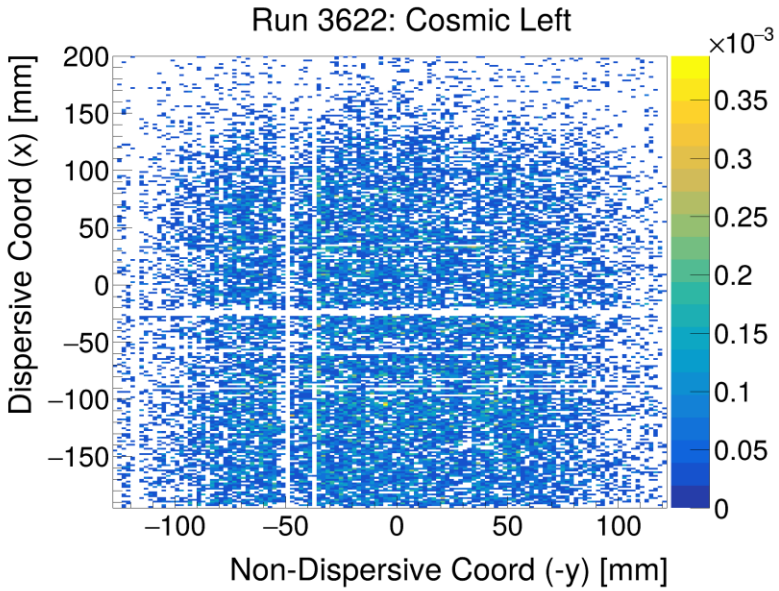


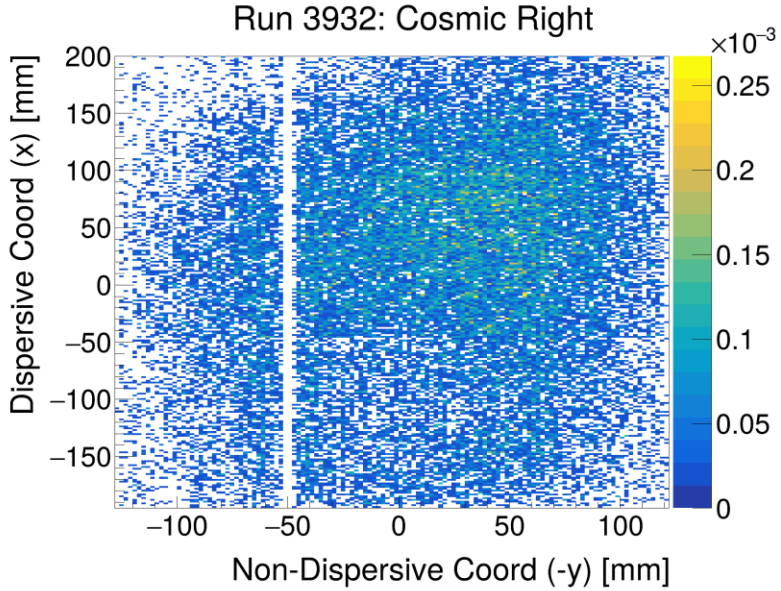


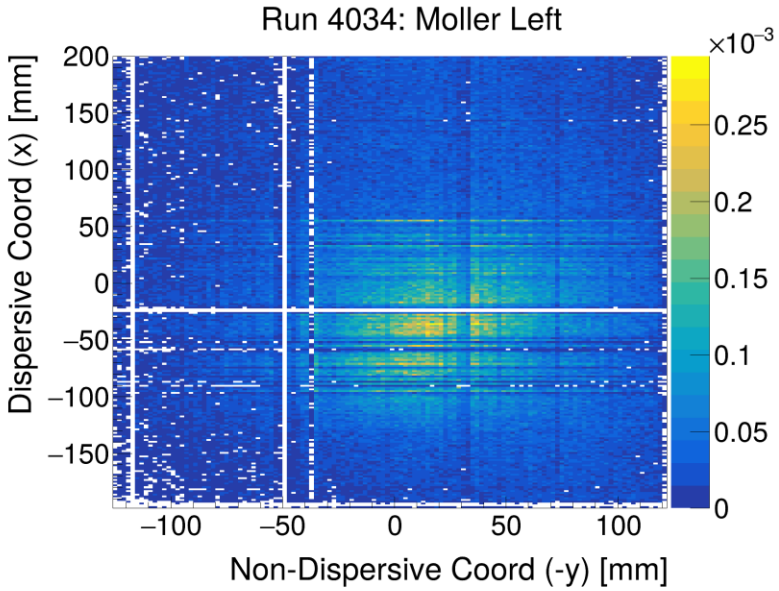


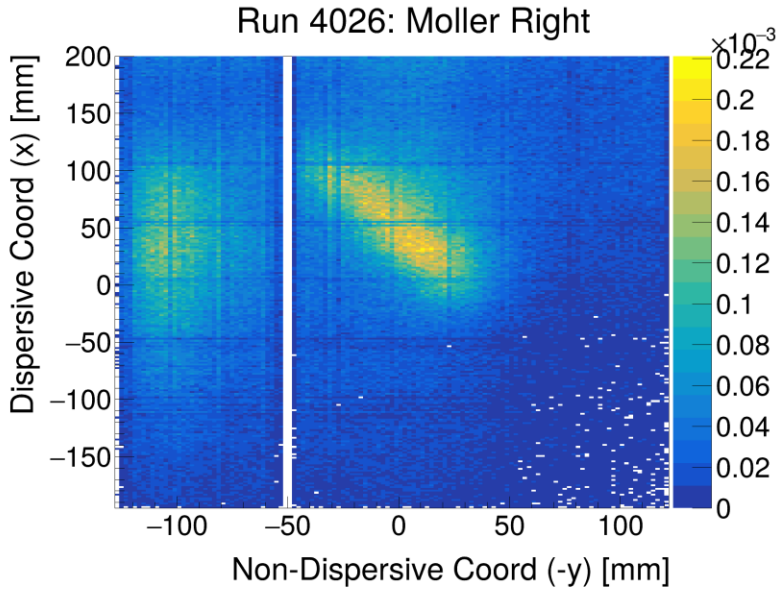


Top GEM Cluster Local Positions



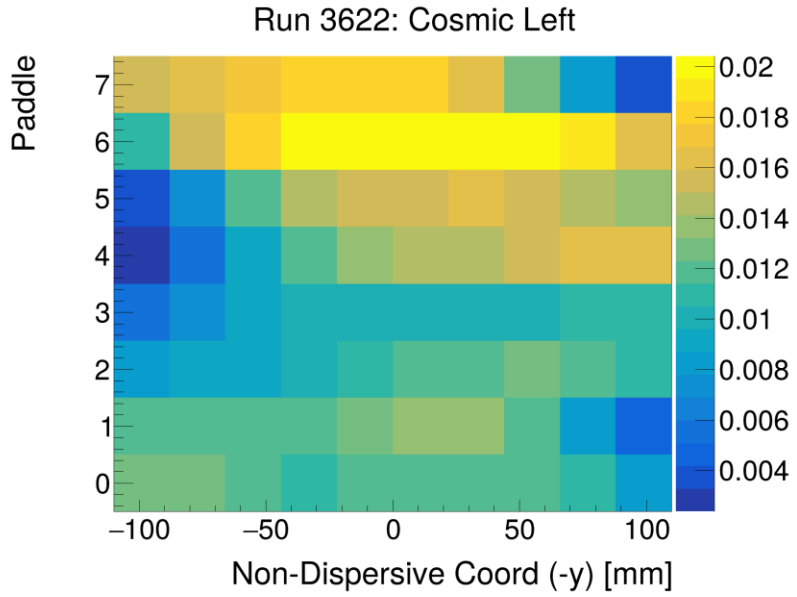


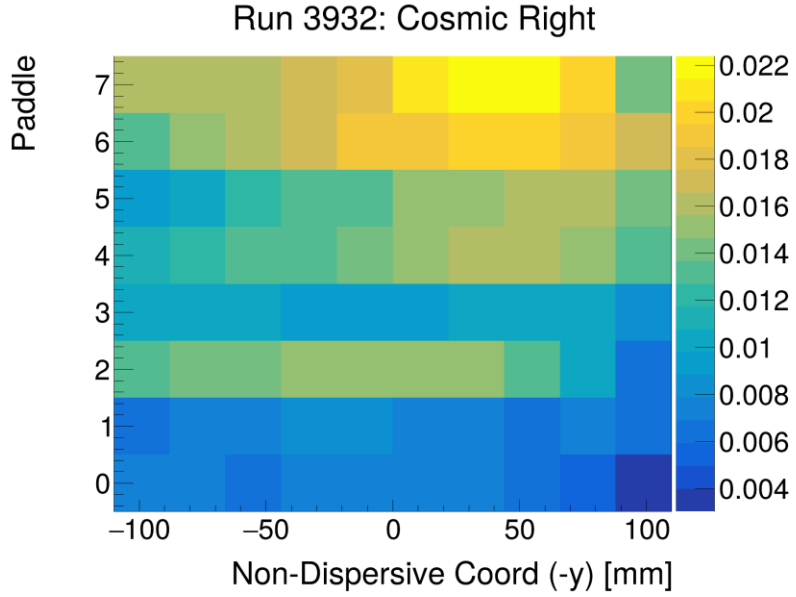


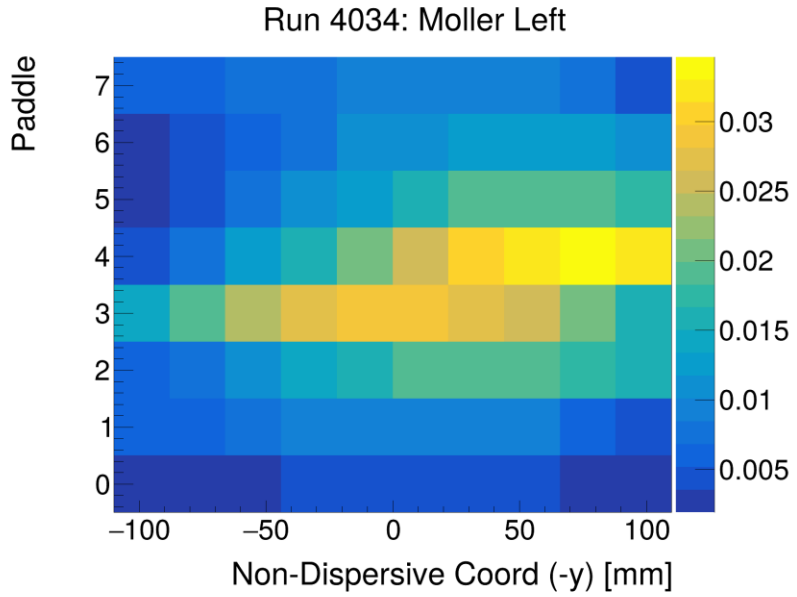


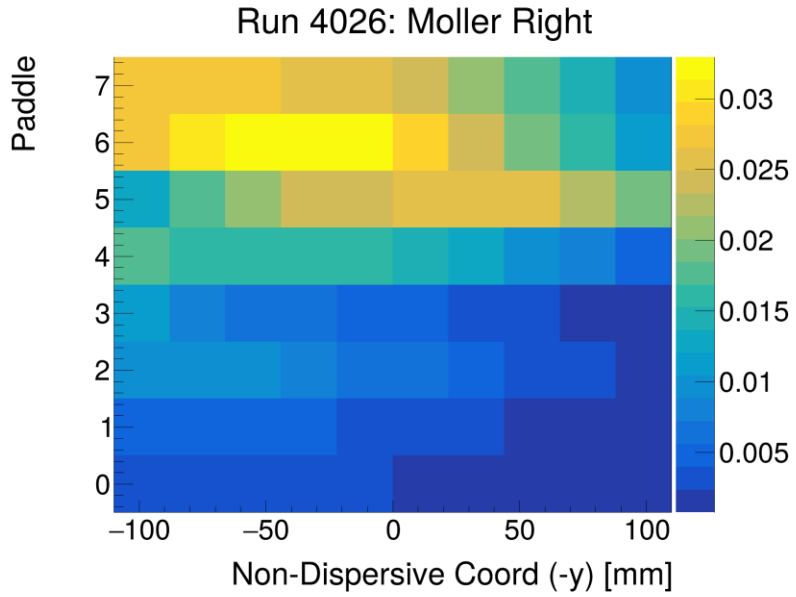
Trigger Hit Positions

Trigger Hit Local Positions



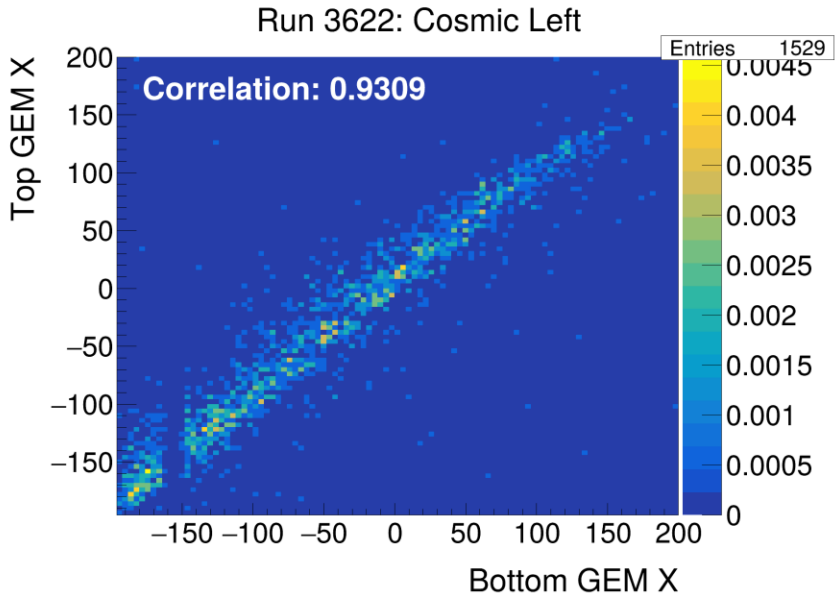


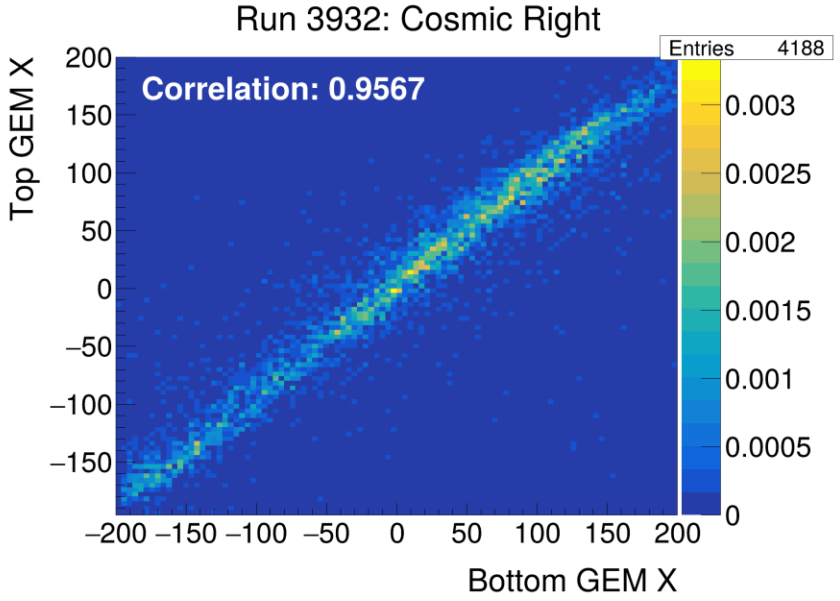


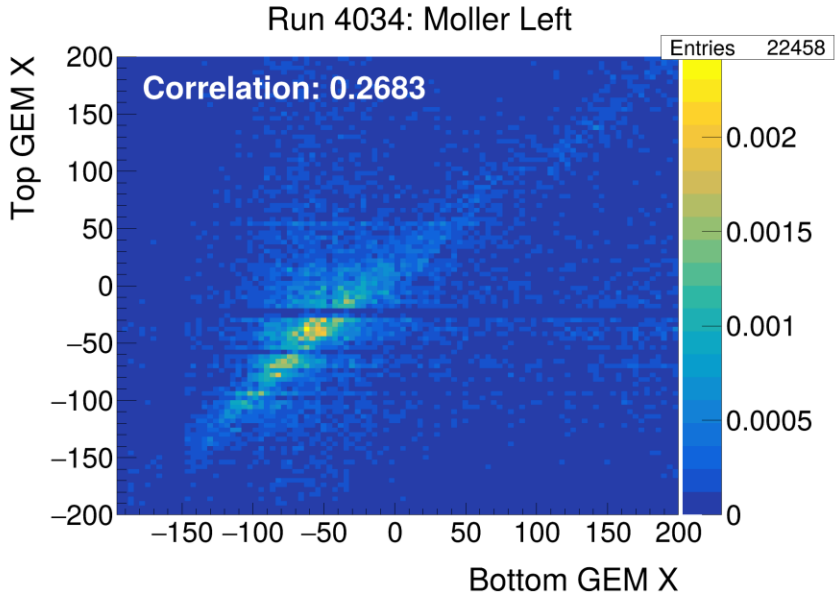


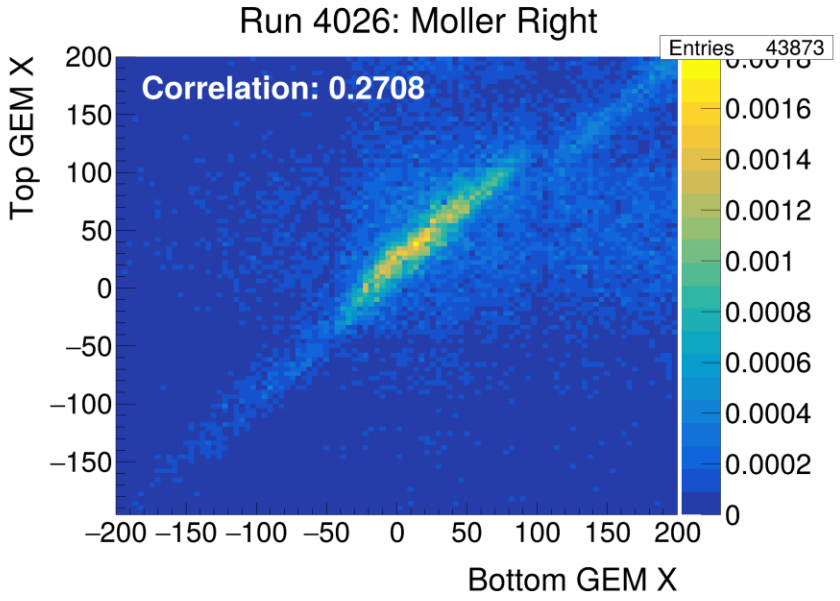
Top vs Bottom GEM Hit Position Correlations

Top GEM vs Bottom GEM Cluster X (Single Hits)

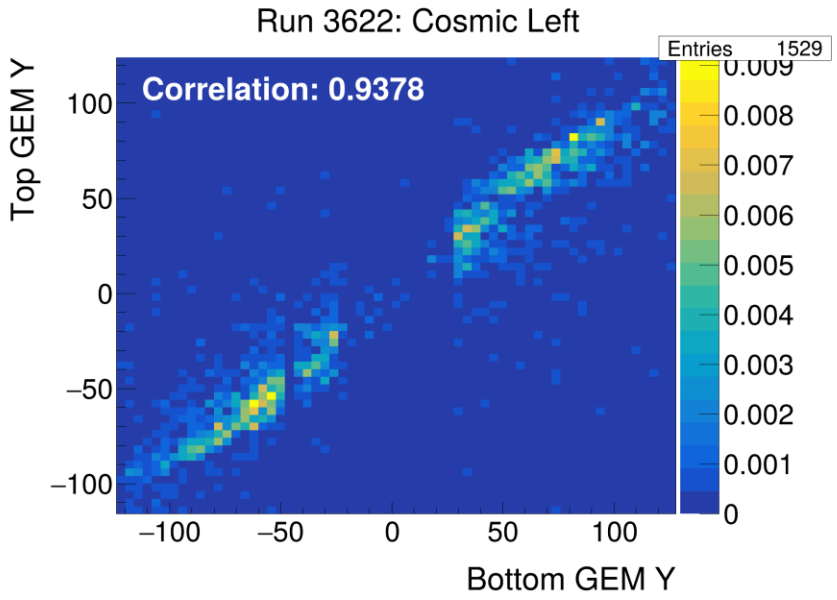


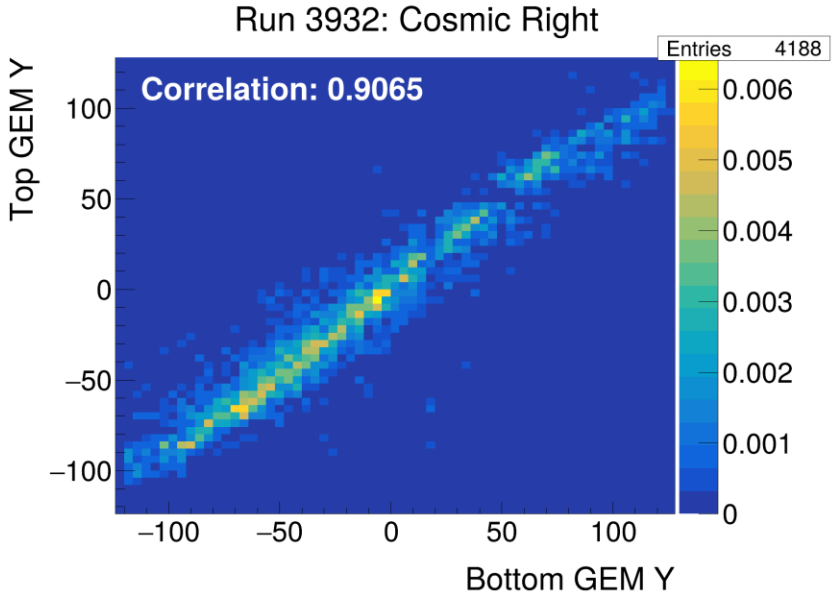


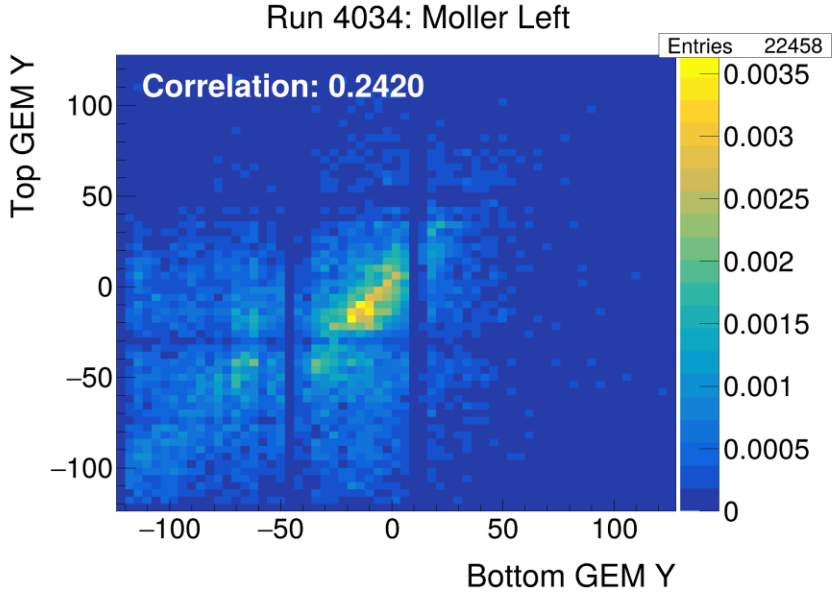


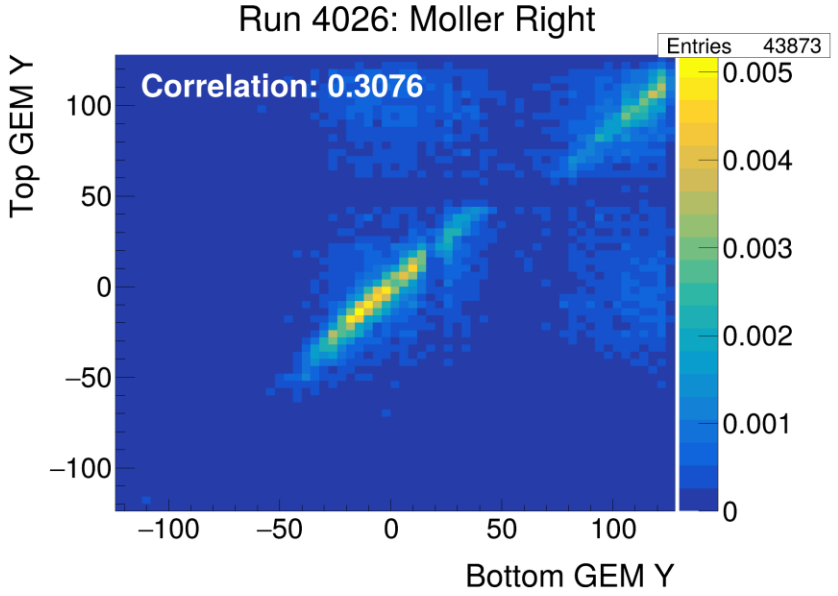


Top GEM vs Bottom GEM Cluster Y (Single Hits)



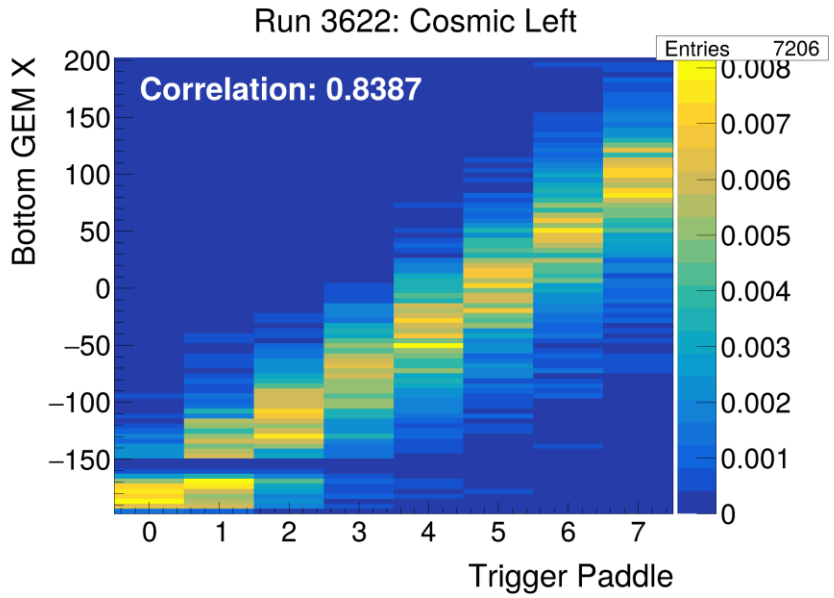


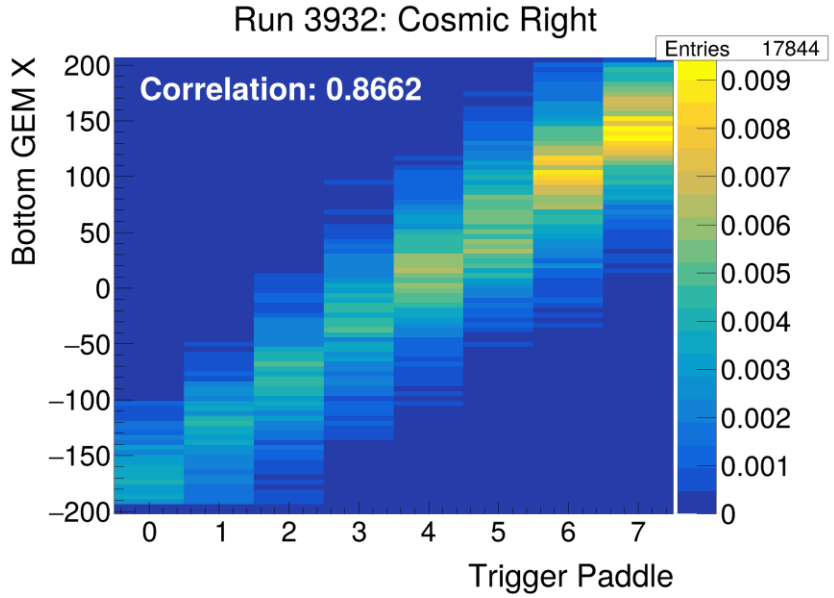


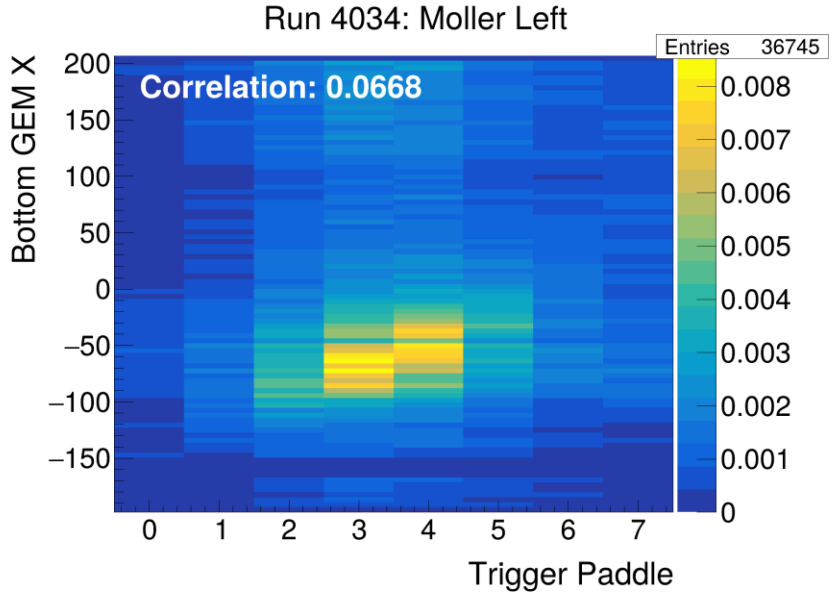


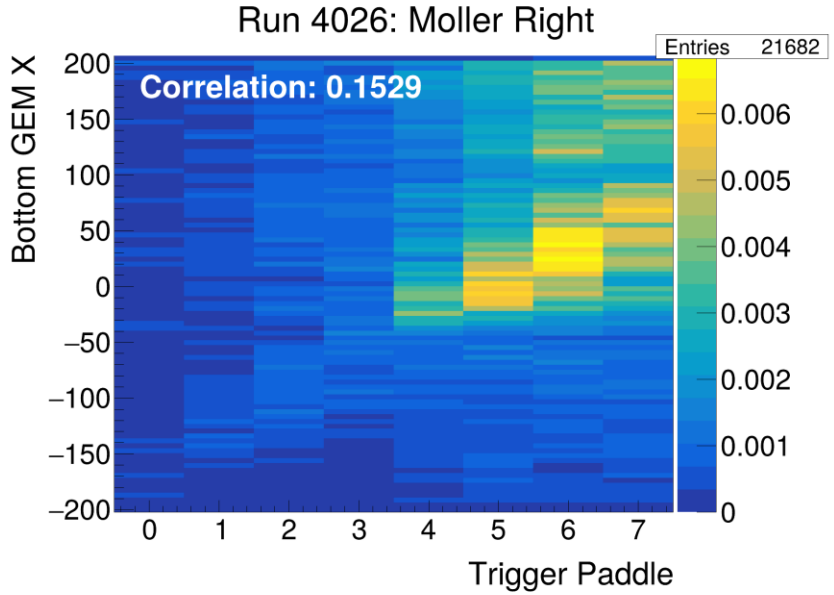
Bottom GEM vs Trigger Hit Position Correlations

Bottom GEM Cluster X vs Trigger Hit Paddle # (Single Hits)

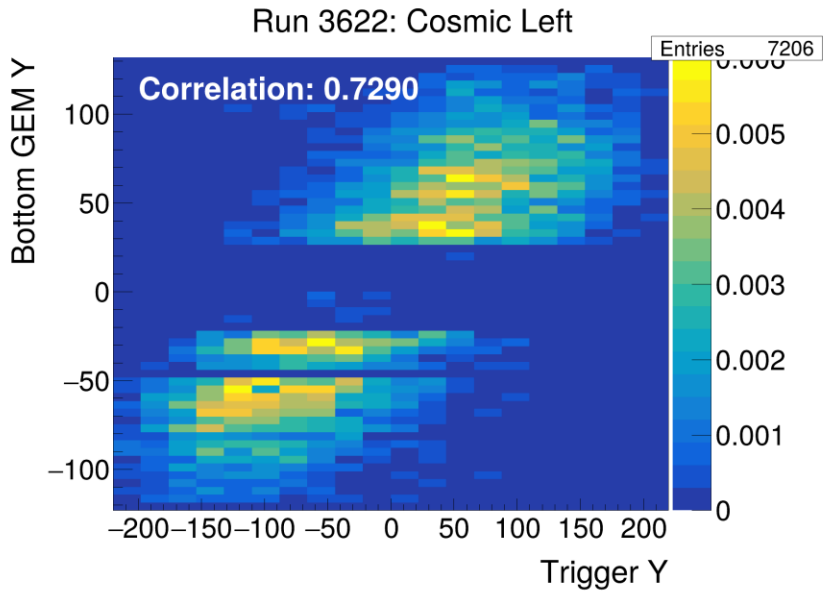


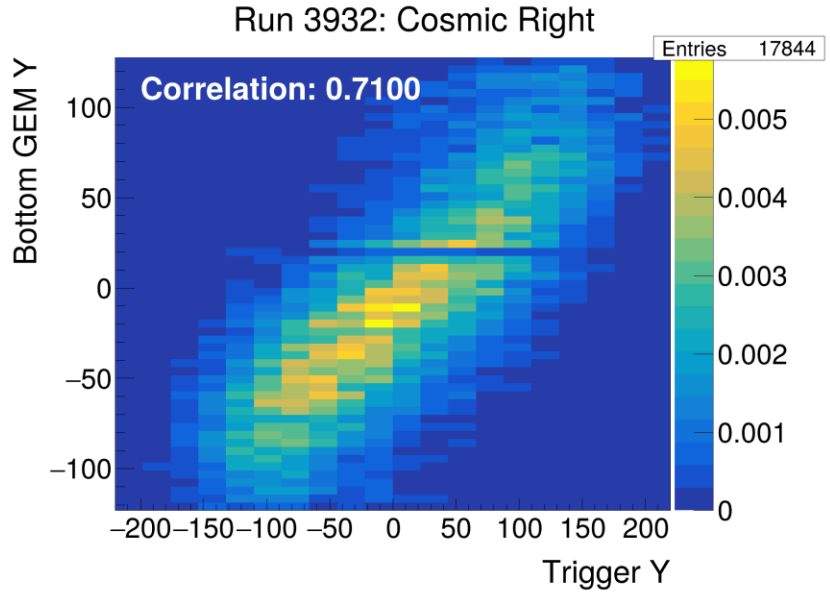


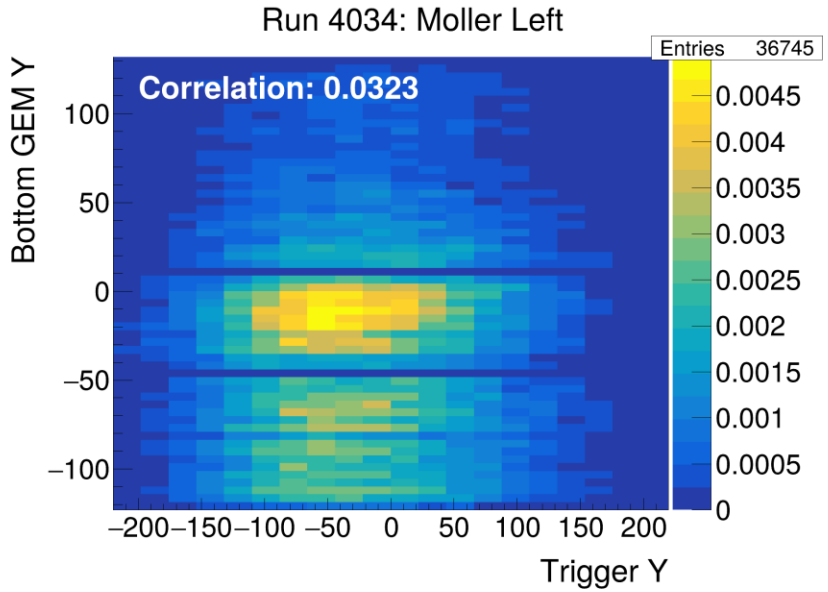


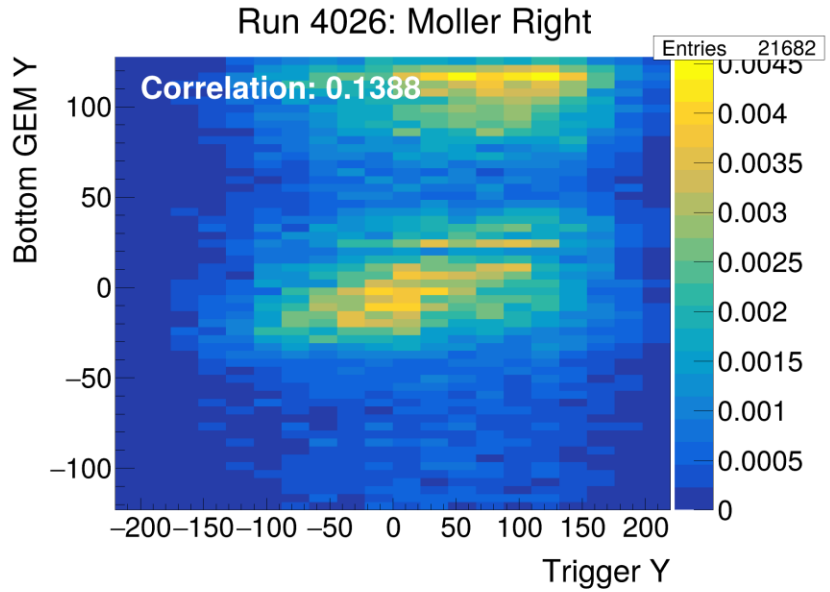


Bottom GEM Cluster Y vs Trigger Hit Y (Single Hits)



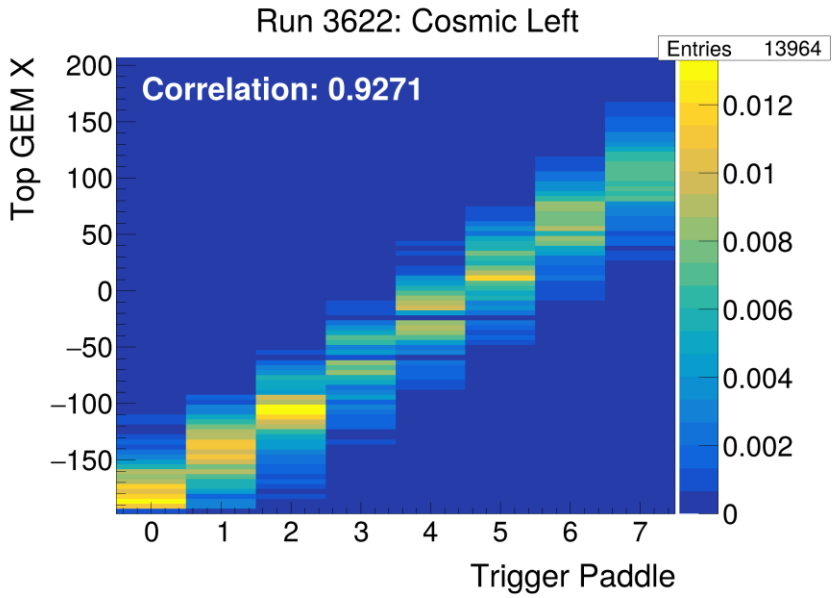


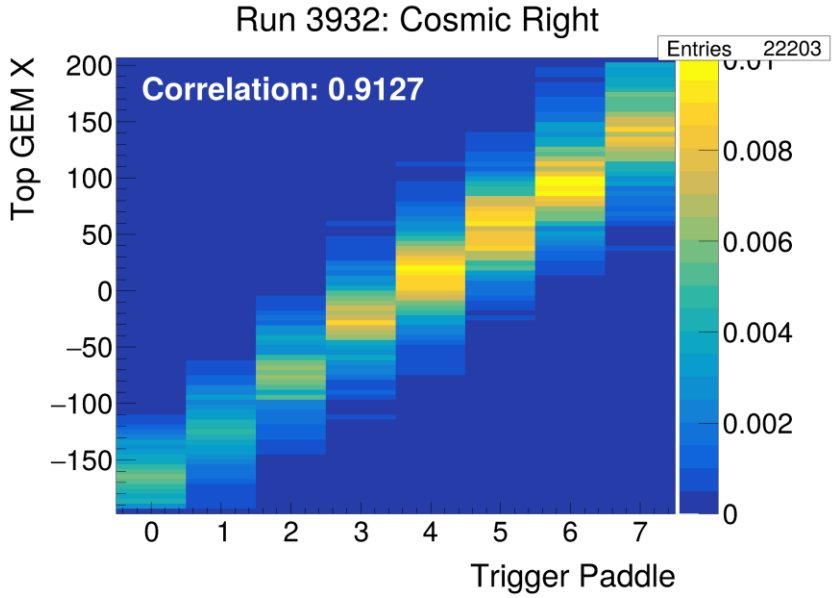


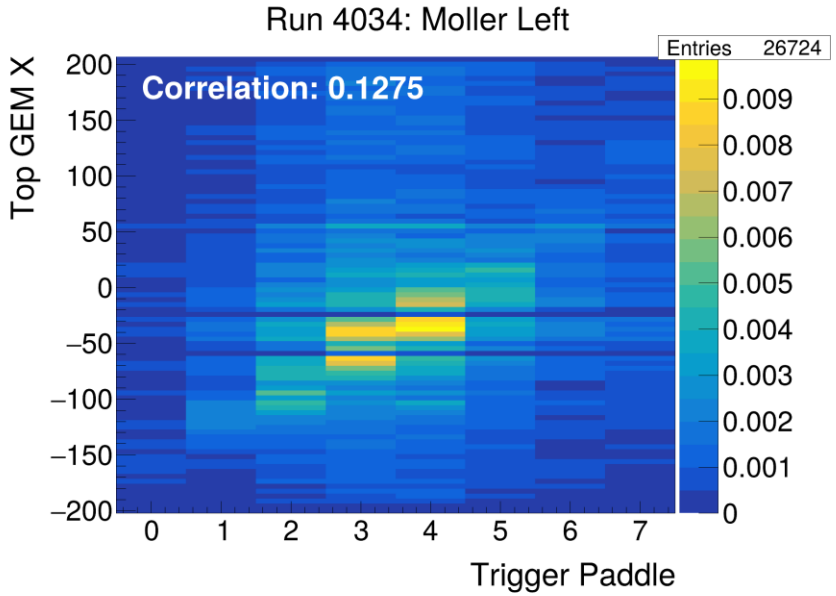


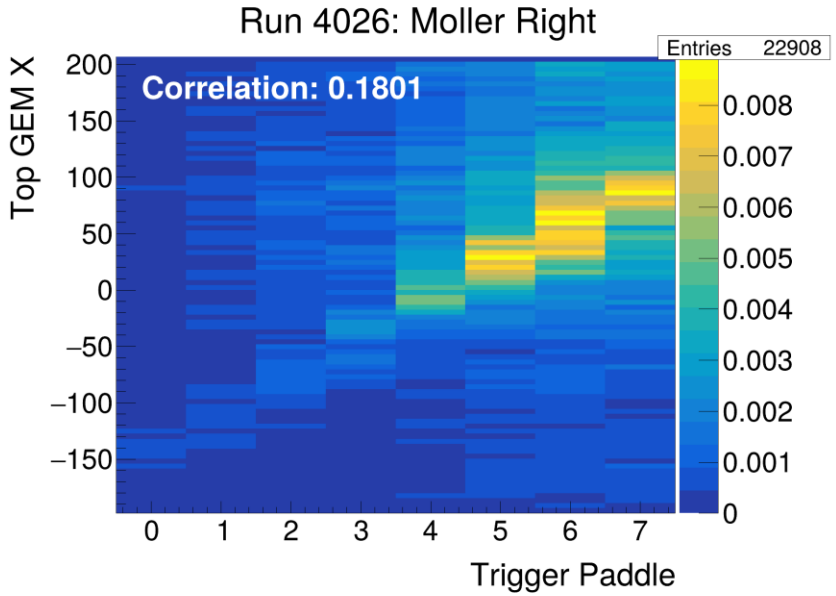
Top GEM vs Trigger Hit Position Correlations

Top GEM Cluster X vs Trigger Hit Paddle # (Single Hits)

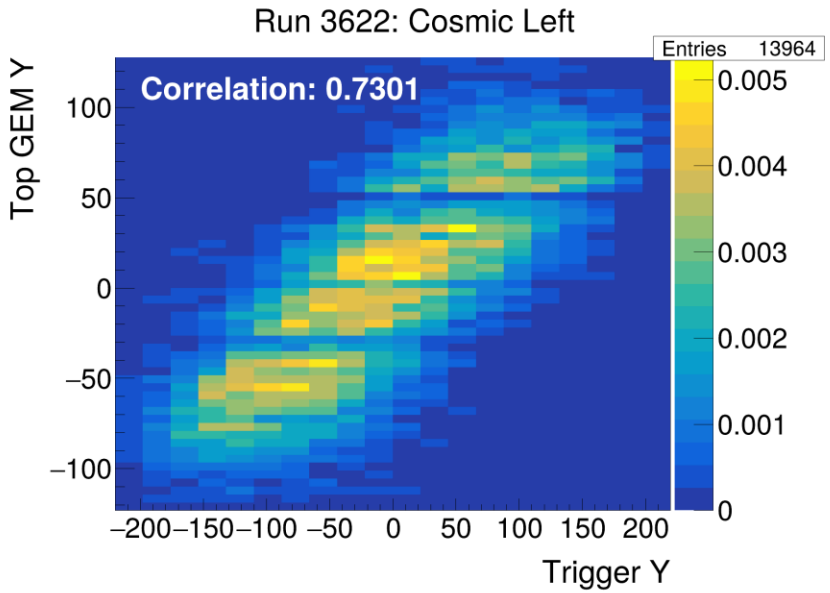


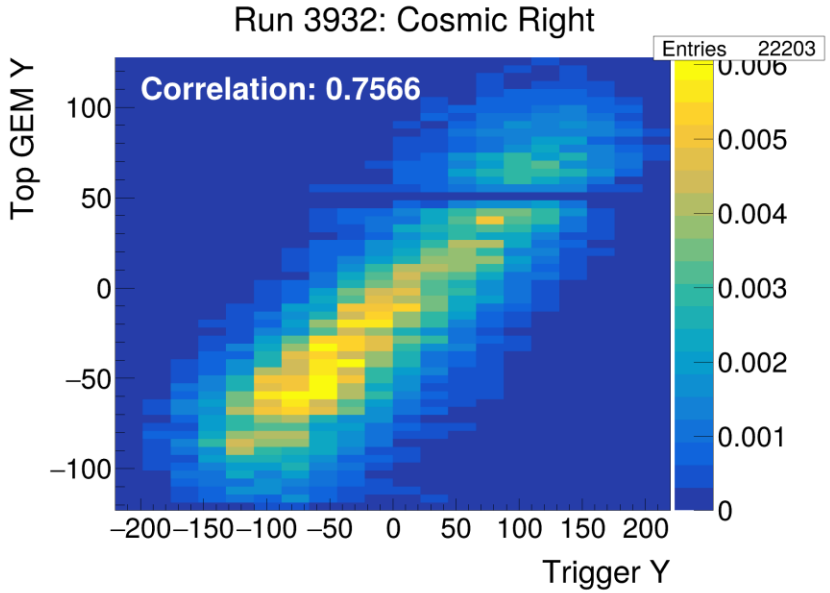


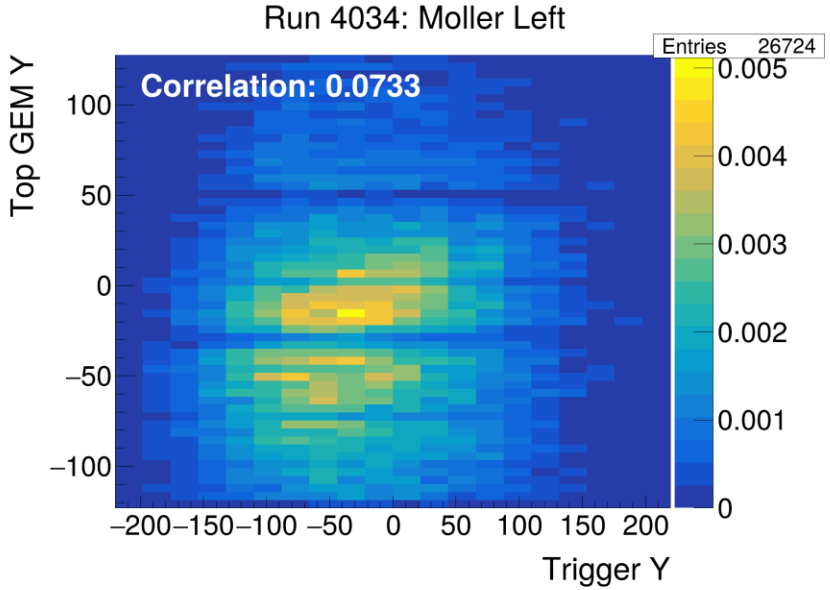


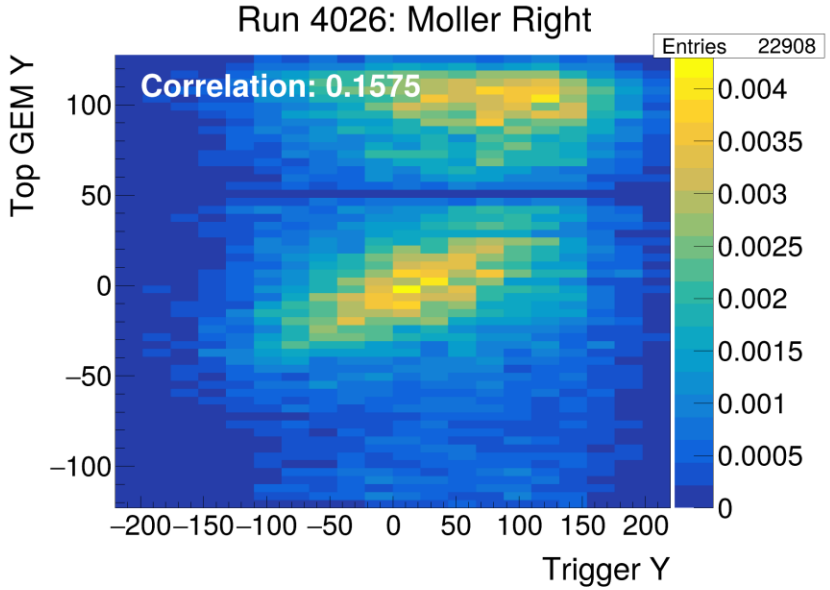


Top GEM Cluster Y vs Trigger Hit Y (Single Hits)



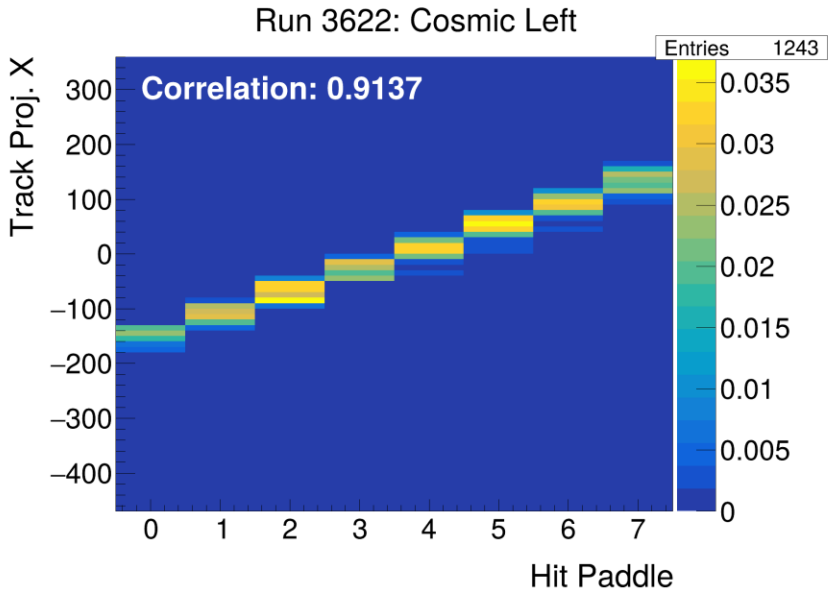


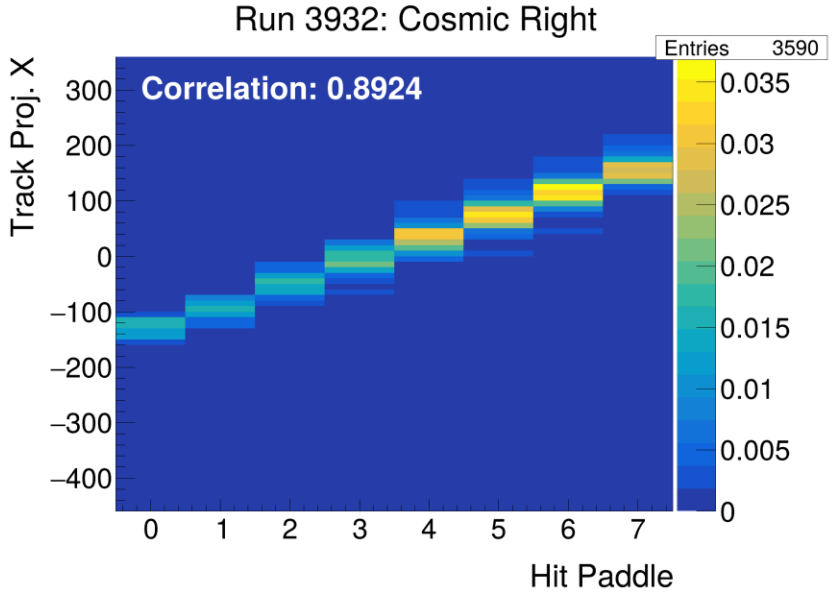


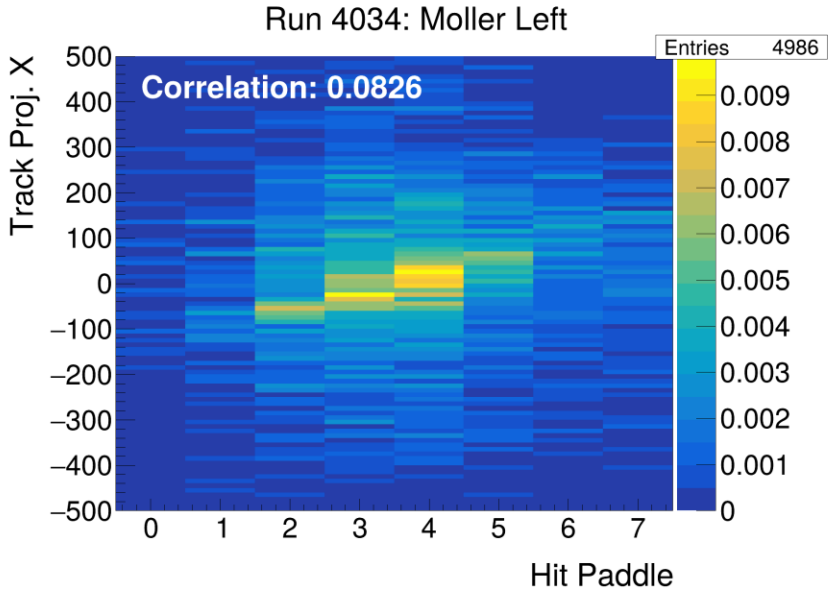


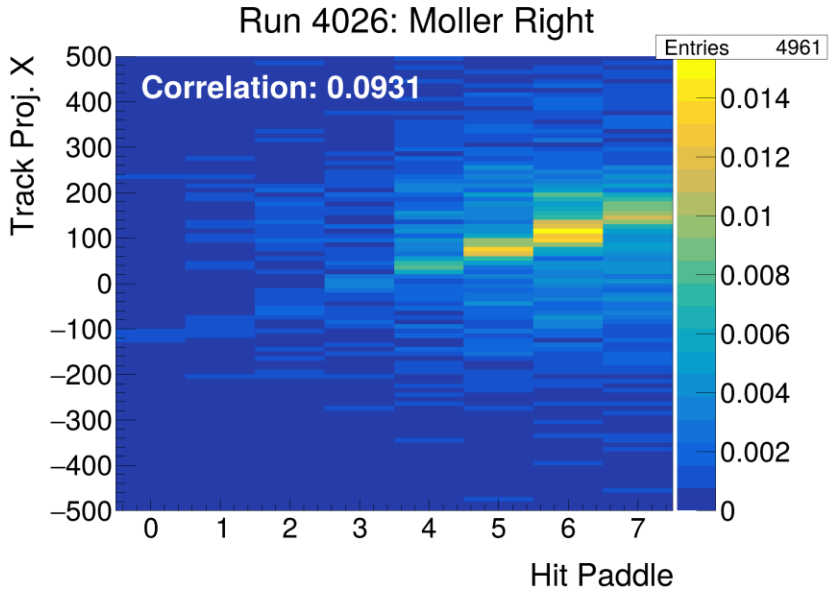
Projected vs Actual Trigger Hit Position Correlations

Track Projection X vs Trigger Hit Paddle # (Single Hits)

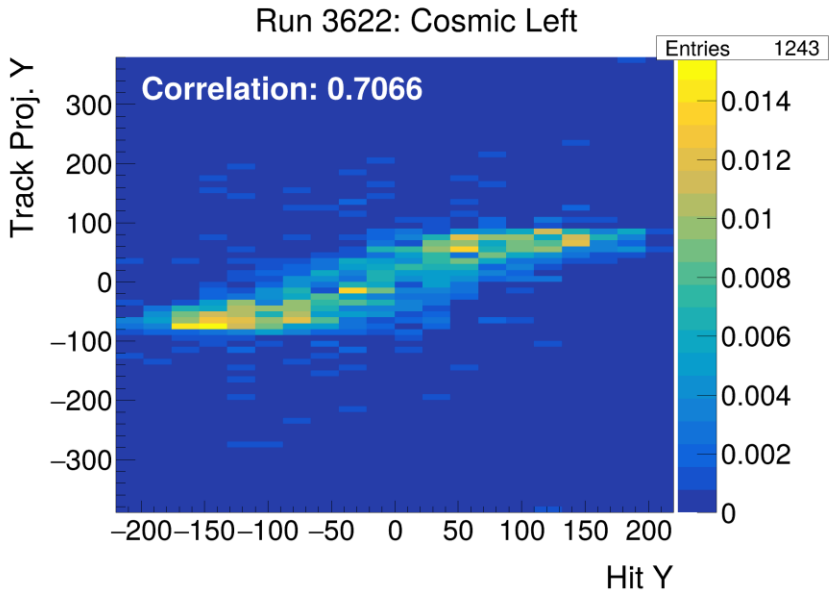


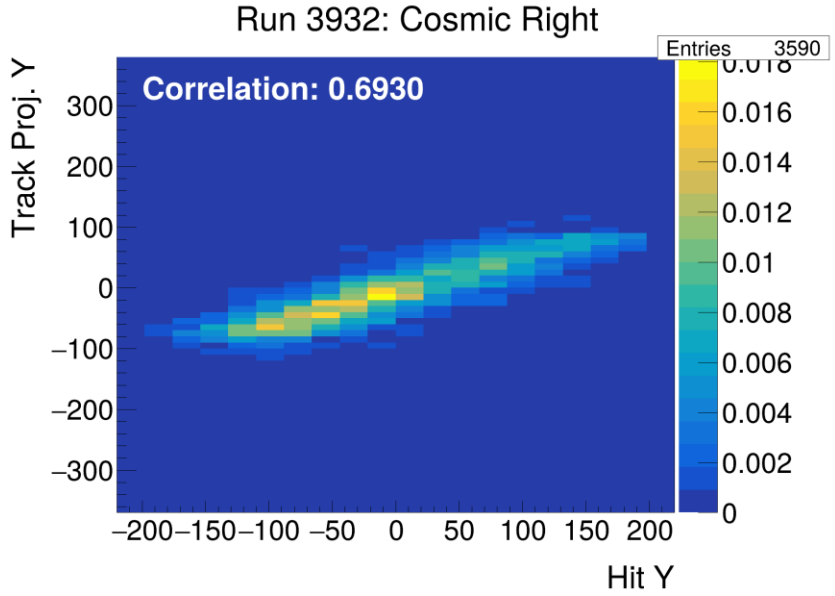


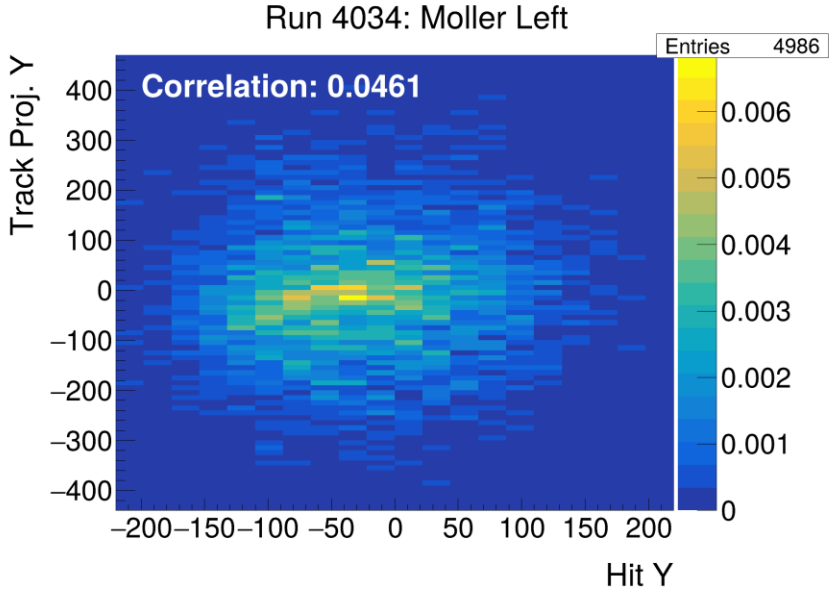


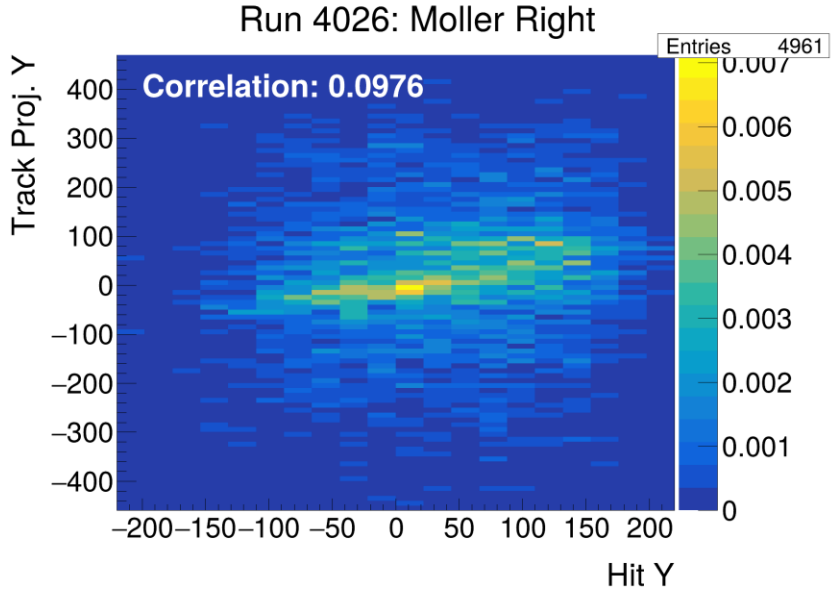


Track Projection Y vs Trigger Hit Y (Single Hits)









Actual vs Projected Hit Paddle Correlations

- If the GEM track projected paddle = -1, then the track projects to a position which does not lie on the trigger paddles

Trigger Hit Paddle # vs Track Projection Paddle # (Single Hits)

